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SRS/SRT SUPPLEMENT

# OPTIMIZING LINAC-BASED STEREOTACTIC RADIOTHERAPY OF UVEAL MELANOMAS: 7 YEARS' CLINICAL EXPERIENCE

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**Purpose: To report on the clinical outcome of LINAC-based stereotactic radiotherapy (SRT) of uveal melanomas. Additionally, a new prototype (hardware and software) for automated eye monitoring and gated SRT using a noninvasive eye fixation technique is described.**

**Patients and Methods: Between June 1997 and March 2004, 158 patients suffering from uveal melanoma were treated at a LINAC with 6 MV (5 × 14 Gy; 5 × 12 Gy prescribed to 80% isodose) photon beams. To guarantee identical patient setup during treatment planning (CT and MRI) and treatment delivery, patients were immobilized with a BrainLAB thermoplastic mask. Eye immobilization was achieved by instructing the patient to fixate on a light source integrated into the mask system. A mini-video camera was used to provide on-line information about the eye and pupil position, respectively. A new CT and magnetic resonance (MR) compatible prototype, based on head-and-neck fixation and the infrared tracking system ExacTrac, has been developed and evaluated since 2002. This system records maximum temporal and angular deviations during treatment and, based on tolerance limits, a feedback signal to the LINAC enables gated SRT.**

**Results: After a median follow-up of 33.4 months (range, 3–85 months), local control was achieved in 98%. Fifteen patients (9.0%) developed metastases. Secondary enucleation was performed in 23 patients (13.8%). Long-term side effects were retinopathy (*n* = 70; 44%), cataract (*n* = 30; 23%), optic neuropathy (*n* = 65; 41%), and secondary neovascular glaucoma (*n* = 23; 13.8%). Typical situations when preset deviation criteria were exceeded were slow drifts (fatigue), large sudden eye movements (irritation), or eye closing (fatigue). In these cases, radiation was reliably interrupted by the gating system. In our clinical setup, the novel system for computer-controlled gated SRT of uveal melanoma was well tolerated by about 30 of the patients treated with this system so far.**

**Conclusion: LINAC-based SRT of uveal melanomas provides good local control. The new prototype system improves the quality of treatment and offers the possibility of movement-gated treatments. In an ongoing study, treatment-related side effects are correlated with dose levels. Such correlations can be used to further optimize linac-based SRT of uveal melanoma.** 

**SRT, LINAC, Conformal RT, Uveal melanoma, Clinical study.**

## INTRODUCTION

The collaborative ocular melanoma study has stated that it is safe to attempt to preserve an eye with a posterior uveal melanoma, because there is no advantage of enucleation over local therapy (1). Melanomas are known to be radioresistant; thus, a high dose is required to achieve tumor control for this type of tumor. For more than 30 years, episcleral plaque therapy has been used successfully to treat posterior uveal melanoma by utilizing the inherent advantage of brachytherapy (i.e., high dose delivered to the target and low doses elsewhere). With this treatment technique, conformal dose distributions are difficult to obtain for tumors near the optic disc and macula and for very large tumors (2, 3). Another highly conformal treatment technique for the management of uveal melanoma is proton beam therapy, which has become the standard external beam therapy technique. Protons have a rapid dose falloff in lateral direction and at the distal part of the beam, which allows sparing of adjacent tissues (4). Besides these two traditional treatment techniques, advanced photon beam therapy techniques have also been applied in studies to explore their potential for the treatment of ocular diseases (5, 6). At the Medical University of Vienna, uveal melanomas have been treated with 3D stereotactic photon beam

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